

of 10,000, the current price. The collateralization ratio is $10,000/9,000 = 111$ percent, which is a margin percentage of 11 percent and is nearly the maximum amount of allowed leverage (10 percent margin).

This intuition works similarly for a short position. The investor has committed to sell at 10,000, which is a positive balance and is supplemented by the margin deposit of 1,000 (so total of 11,000). The investor's negative balance is the obligation to buy 1 BTC, currently worth 10,000. The collateralization ratio is $11,000/10,000$, which corresponds to a margin of 10 percent.

Let's now follow the mechanics of a short position when the underlying asset (BTC) increases in value by 5 percent. If the price of BTC increases to 10,500 (a 5 percent increase), the margin percentage becomes $(11,000/10,500) - 1 = 4.76$ percent and the short position becomes subject to liquidation. The net balance of the position is \$500, the incentive for the liquidator to close the position collect the balance. Figure 6.10 reviews the mechanics of a long position.

The dYdX BTC perpetual futures contract allows investors to access BTC returns natively on the Ethereum blockchain while being able to supply any ERC-20 asset as collateral. Perpetual futures are rising in popularity, and this functionality may continue to attract liquidity over time.